

Cyclomatic Complexity using RAPTOR

Aim: To calculate the Cyclomatic Complexity of a flow graph using the given values of edges, nodes, and predicate nodes.

Theory:

Cyclomatic Complexity is a software testing metric used to measure the complexity of a program. It indicates the number of linearly independent execution paths in a program. It can be calculated using:

$$V(G) = E - N + 2$$

or

$$V(G) = P + 1$$

where E = Number of Edges, N = Number of Nodes, and P = Number of Predicate Nodes.

Given Values:

Edges (E) = 17

Nodes (N) = 13

Predicate Nodes (P) = 5

Calculation:

Using $V(G) = E - N + 2$:

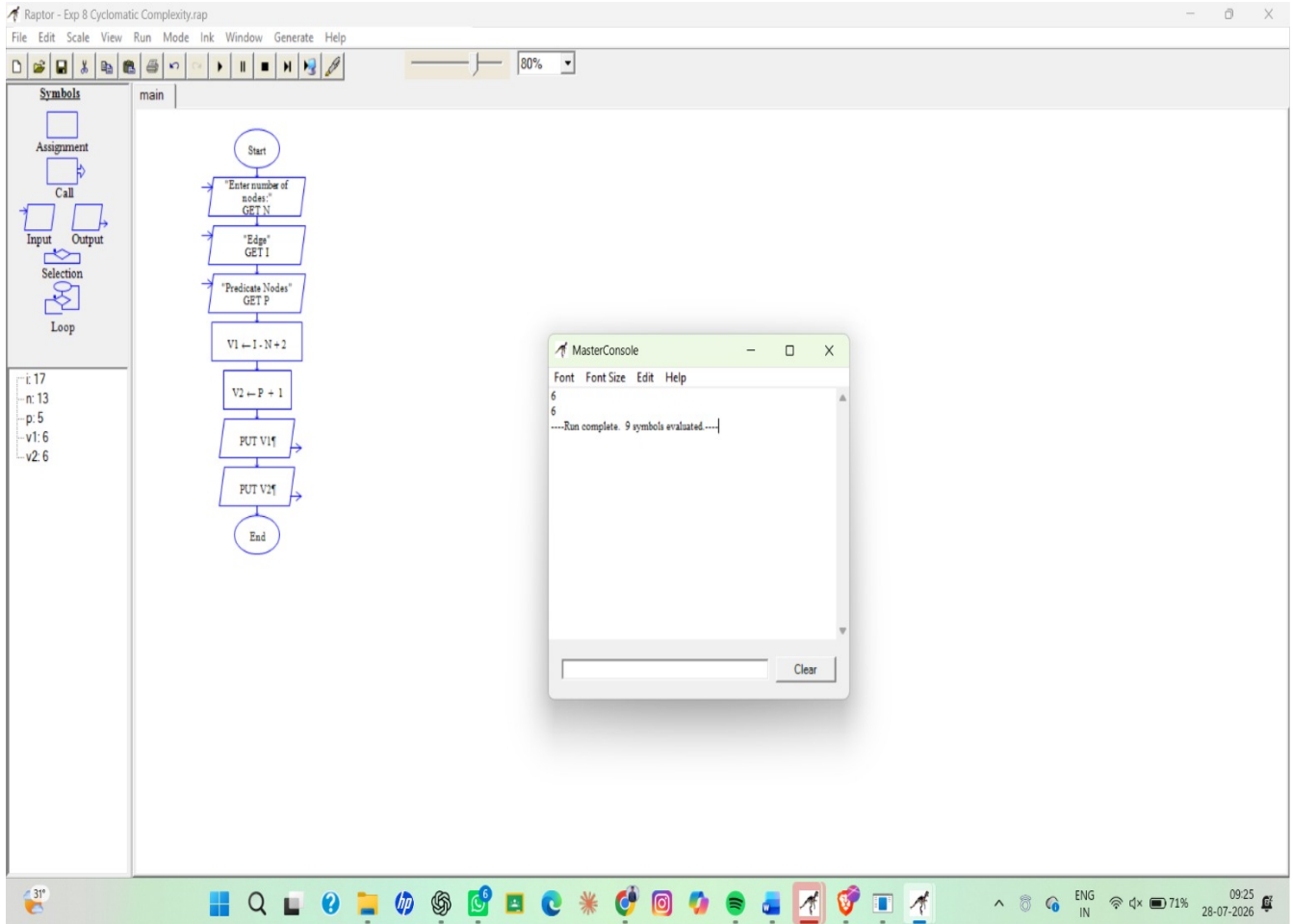
$$V(G) = 17 - 13 + 2 = 6$$

Using $V(G) = P + 1$:

$$V(G) = 5 + 1 = 6$$

Hence, the Cyclomatic Complexity is **6**.

RAPTOR Flowchart:



Result: The RAPTOR program successfully calculated the Cyclomatic Complexity. Both formulas produced the same result, therefore the Cyclomatic Complexity of the given flow graph is **6**.